Methods description — MVA Hackathon 2026, Track 1 (Variant Prediction)

Please submit your answers with your Track 1 submission. All questions (with the exception of Question 1, in red) are voluntary; how
since the panel can't assess what it can't see.

Fill out this form once per model/approach submitted.

Team name

Model number (fill out this form once per model/approach submitted; up to 6)

Please describe your model/approach in detail.

(required) If commercially-available Generative AI was used, record the provider, the plan or tier, and the relevant setting in your methods description. A line is enough, for example, "Anthropic API, Claude Sonnet, commercial terms, no training on customer content."

Please state whether the submission file is the automated output of your computational approach (preferred), or if it has undergone downstream manual review and curation.

Please describe any downstream manual review in detail.

Please state whether your approach only used publicly available data (preferred) or if proprietary data was also used. If proprietary data were used, where possible, please also include a submission based only on publicly available data.

Please describe any public data used in detail (e.g., reference panels, population databases, literature).

Please describe any proprietary data used in detail.

Is your approach able to output proposed pairs of compound heterozygous candidate variants, or only single candidate variants?

How did you handle secondary or incidental findings, if any? (See the finding_type column in your submission.)

Please provide an estimate of run time and cost for your approach (if possible).

Provide a method abstract (up to 500 words). Please include strengths and limitations of your method as well as methodological detail.

ever, submissions without a methods description may be judged less favorably on the Innovation and Scalability criteria,

Answers

RareELleles

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Six independent prioritisation approaches were run on the proband's WGS and their outputs integrated into a single ranked list.
1. Alignment and QC: reads aligned to GRCh38 (no-alt + hs38d1, GIAB v2 mask, matching the provided VCF) with bwa mem 0.7.19; duplicates marked with GATK MarkDuplicates; mean depth about 40x.
2. Small variants: DeepVariant 1.10.0 calls plus the provided Sentieon VCF (genotype concordance 0.994), annotated with VEP 116.2 (gnomAD v4.1, AlphaMissense) and ClinVar, and phased with WhatsHap 2.8.
3. Structural variants: Cue and Manta, cross-confirmed with truvari and annotated with gnomAD-SV and AnnotSV 3.5; exon read depth screened over the MVA genes.
4. Ranking:
(a, b) Exomiser 15.1.0 on the DeepVariant and the provided call sets (two lists), with the HPO terms from the provided phenotype document;
(c) a tiered rule-based ranking combining small variants, SVs, phasing, Exomiser and AlphaGenome;
(d) a ranking of rare variants by AlphaGenome Atlas (AVI) score;
(e) AlphaGenome 0.9.0 scoring of rare variants within 50 kb of 11 MVA marker genes;
(f) ACMG point-based scoring of marker-gene variants and nearby SVs.
5. Consensus: the top 30 of each list were matched by variant and given six evidence flags (MVA gene, two or more independent methods, protein-altering or SV, rare, ClinVar/AlphaMissense pathogenic, fits recessive inheritance). All six lists ranked BUB1B p.Leu737Ter and p.Asn1002Lys at the top.

Anthropic Claude (Sonnet and Opus models), Elucidata business plan (Team/Enterprise), commercial terms, no training on customer content; no feedback or ratings were given on outputs. Used for writing and checking analysis scripts, summarising and cross-checking result tables, and drafting documentation.

The submission file has undergone downstream manual review and curation. The candidate lists are the automated outputs of six pipelines; the submitted rows, their order and EPCR values were set by manual review of those outputs, as described below.

- Compared the top 30 of the six rankings (148 distinct variants, 126 of them in only one list) and the SV lists.
- Reviewed an evidence table with six flags per variant and warnings for ClinVar benign, AF of 1% or more, cis pairs and annotation errors.
- Checked top candidates in the BAM (both call sets, allele counts, strand, MAPQ). For example, a ClinVar P/LP call in PRSS1 was set aside: DeepVariant did not call it and all alternate reads at MAPQ 20 or more were on one strand.
- Checked WhatsHap phase for same-gene pairs: the ADAMTS1 and FANCD2 pairs are in cis; the BUB1B pair (about 11 kb apart) could not be phased.
- Assigned EPCR values in descending order of supporting evidence: the BUB1B compound-heterozygous pair, then each BUB1B variant alone, then rare non-coding variants near MVA marker genes, then rare predicted-damaging variants in other genes; incidental findings received the lowest values.

Only publicly available data were used. Apart from the dataset provided by the organisers (WGS FASTQs, VCF and phenotype document), all resources were public: the GRCh38 reference and GIAB masks, Ensembl and MANE gene models, gnomAD (v4.1 and gnomAD-SV), ClinVar, AlphaMissense, ReMM and the Exomiser data release, UCSC centromere models, the AlphaGenome model and AlphaGenome Atlas, and published MVA gene literature. No proprietary data were used for Track 1.

Reference: GRCh38 NCBI no-alt analysis set with hs38d1 decoys and the GIAB v2 masked exclusions. Gene models: Ensembl release 116 GTF and MANE Select v1.5. Population frequencies: gnomAD v4.1 (VEP 116 cache) and gnomAD-SV v4.1 (via SVAFotate).

None

Yes. Compound-heterozygous pairs are proposed when a gene carries two rare, predicted-damaging heterozygous variants (Exomiser autosomal recessive model and the two-hit tiers of the rule-based ranking). WhatsHap phase sets are used to exclude

Two incidental findings are reported with finding_type secondary: LZTR1 p.(Tyr748Ter) (heterozygous stop-gain, ClinVar pathogenic/likely pathogenic) and GNRHR p.(Arg139His) (heterozygous, ClinVar pathogenic; carrier finding in an autosomal

All analyses ran on a single cloud server: Intel Core i9-14900K (1 socket, 32 logical CPUs), 125 GiB RAM, 1.8 TB disk, Ubuntu 24.04.5 LTS, no GPU. Approximate compute time per stage: reference build, alignment, QC, DeepVariant calling, phasing, Exomiser and annotation about 14 hours (alignment 4.4 hours and DeepVariant 3.5 hours are the longest steps); structural-variant calling with Cue and Manta about 5 hours; structural-variant annotation about 1 hour; Exomiser on the provided calls about 40 minutes; AlphaGenome marker-gene scoring 1.5 to 3 hours; AlphaGenome Atlas ranking about 2 hours; final ranking 1 to 2 hours. Run one after another, this is about 25 to 27 hours of compute; several stages can run in parallel. Compute cost was about $225.

We prioritised candidate causal variants for the proband from whole-genome sequencing by combining independent variant calling, phasing, structural-variant detection and six ranking approaches, then selected the submitted rows by manual review of cross-method agreement and supporting evidence.

Methods. Reads were aligned to GRCh38 (NCBI no-alt analysis set with hs38d1 decoys and the GIAB v2 mask, matching the provided VCF) with bwa mem and duplicate-marked with GATK; sample identity was confirmed against the provided VCF (concordance 0.994). Small variants were called with DeepVariant 1.10.0 and analysed alongside the provided Sentieon calls; both sets were normalised, filtered to PASS, annotated with Ensembl VEP 116.2 (gnomAD v4.1, AlphaMissense) and ClinVar, and read-backed phased with WhatsHap. Structural variants were called with Cue and Manta, cross-confirmed with truvari and annotated with gnomAD-SV and AnnotSV, and exon read depth was screened over the MVA genes. Six lists were produced: Exomiser on each call set with the provided HPO terms; a tiered rule-based ranking that integrates small variants, SVs, phase, Exomiser and AlphaGenome and puts biallelic genotypes in MVA genes first; a rare-variant ranking by AlphaGenome Atlas (AVI) score; AlphaGenome scoring of rare variants around 11 MVA genes; and ACMG point-based scoring of marker-gene variants and SVs. The top 30 of each list were matched by variant and annotated with six evidence flags.

Results. All six lists placed BUB1B c.2210T>G p.(Leu737Ter) and c.3006T>G p.(Asn1002Lys) at the top. Below these, agreement was low: 148 distinct variants, 126 of them in a single list.

Strengths: two independent call sets; methods with different assumptions (phenotype-driven, rule-based, regulatory prediction, ACMG-style scoring); SV and exon CNV screening; read-level and phase review; and only public resources with versioned software environments.

Limitations: a single proband without parental samples, so the two BUB1B variants (about 11 kb apart) could not be phased and their trans configuration is unconfirmed; CADD was not available, and REVEL and SpliceAI scores were available only through Exomiser; depth-based exon CNVs and single-caller SVs are unconfirmed; distal regulatory predictions are less reliable; the chromosome-level depth screen cannot detect aneuploidy in fewer than about 15% of cells; and the final row selection, order and EPCR values were set by manual curation rather than a fully automated score.